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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 1st September Evening Shift

JEE Main Question Paper with Answer Key

1 Sep 2021	90	16	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	23 questions
Mathematics	23 questions
Physics	28 questions
Out of syllabus	16 questions
Answer key	at the end

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Chemistry 23 questions

Q1. MCQ (Single Correct Answer)

Experimentally reducing a functional group cannot be done by which one of the following reagents?

(A) $Pt\text{-}C/H_2$ (B) Na/H_2 (C) $Pd\text{-}C/H_2$ (D) Zn/H_2O

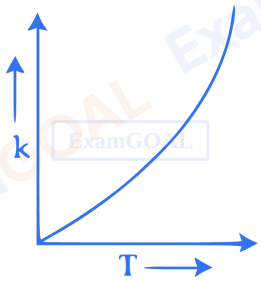
Open solution

Q2. MCQ (Single Correct Answer)

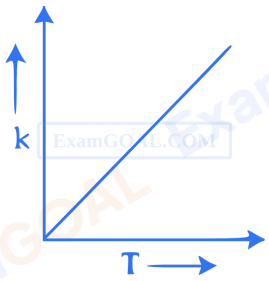
Which one of the following given graphs represents the variation of rate constant (k) with temperature (T) for an endothermic reaction?

Open solution

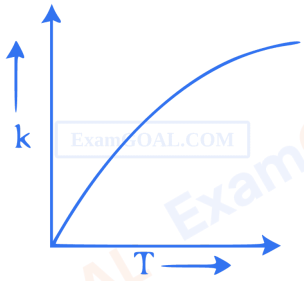
(A)



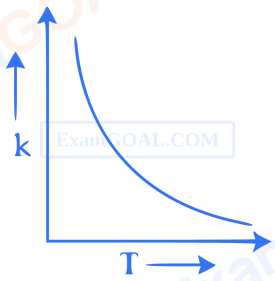
(B)



(C)



(D)

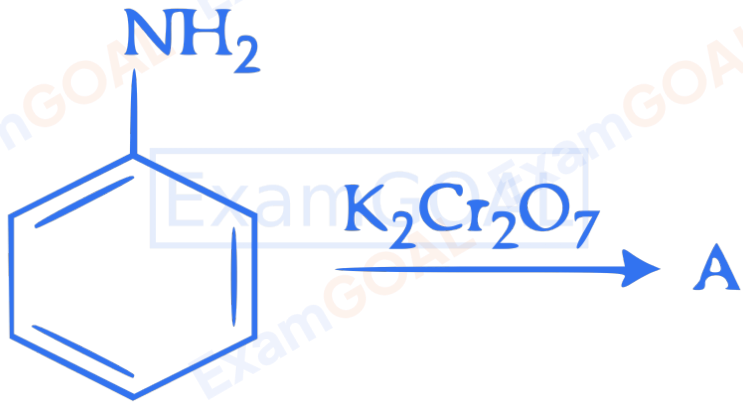


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Q3. MCQ (Single Correct Answer)

Identify A in the following reaction.

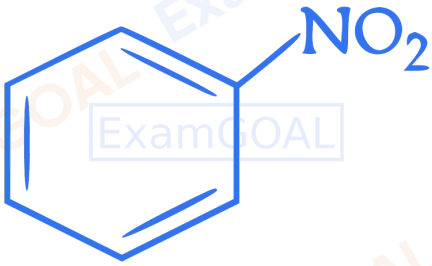
Open solution



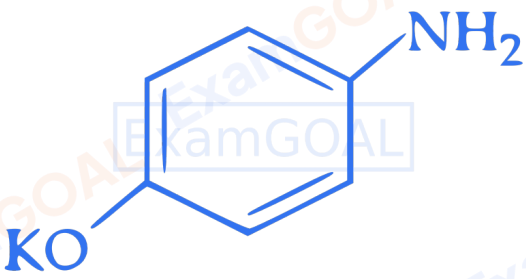
(A)



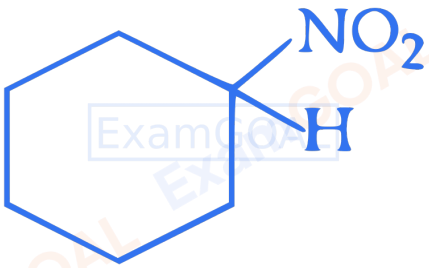
(B)



(C)



(D)



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Q4. MCQ (Single Correct Answer)

In the following sequence of reactions a compound A, (molecular formula $C_6H_{12}O_2$) with a straight chain structure gives a C_4 carboxylic acid. A is :

Open solution

$$\mathbf{A} \xrightarrow[H_3O^+]{LiAlH_4} \mathbf{B} \xrightarrow{\text{Oxidation}} C_4 \text{ - carboxylic acid}$$

(A) $CH_3 - CH_2 - COO - CH_2 - CH_2 - CH_3$

(B)



(C) $CH_3 - CH_2 - CH_2 - COO - CH_2 - CH_3$

(D) $CH_3 - CH_2 - CH_2 - O - CH = CH - CH_2 - OH$

Q5. MCQ (Single Correct Answer)

The Crystal Field Stabilization Energy (CFSE) and magnetic moment (spin-only) of an octahedral aqua complex of a metal ion (M^{z+}) are $-0.8\ \Delta_0$ and 3.87 BM, respectively. Identify (M^{z+}) :

Open solution

(A) V^{3+} (B) Cr^{3+} (C) Mn^{4+} (D) Co^{2+}

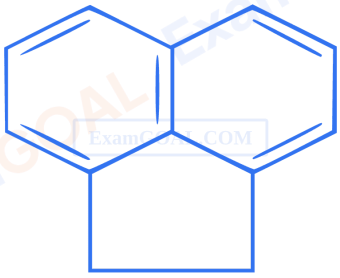
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Q6. MCQ (More than One Correct Answer)

Open solution

Which one of the following compounds is aromatic in nature?

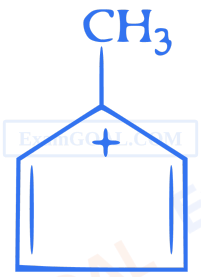
(A)



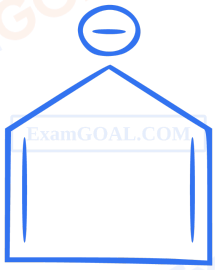
(B)



(C)



(D)



Q7. MCQ (Single Correct Answer)

Open solution

In the given chemical reaction, colors of the Fe^{2+} and Fe^{3+} ions, are respectively :

$$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$$

(A) Yellow, Orange (B) Yellow, Green (C) Green, Orange (D) Green, Yellow

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Q8. MCQ (Single Correct Answer)

The stereoisomers that are formed by electrophilic addition of bromine to trans-but-2-ene is/are :

(A) 2 enantiomers and 2 mesomers

(B) 2 identical mesomers

(C) 2 enantiomers

(D) 1 racemic and 2 enantiomers

Open solution

Q9. MCQ (Single Correct Answer)

In the following sequence of reactions,



The compounds B and C respectively are :-

(A) Cl_3COOK, HCOOH (B) Cl_3COOK, CH_3I (C) CH_3I, HCOOK (D) CHI_3, CH_3COOK

Open solution

Q10. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : The nucleophilic addition of sodium hydrogen sulphite to an aldehyde or a ketone involves proton transfer to form a stable ion.

Statement II : The nucleophilic addition of hydrogen cyanide to an aldehyde or a ketone yields amine as final product.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement are true.

(B) Statement I is true but Statement II is false.

(C) Statement I is false but Statement II is true.

(D) Both Statement I and Statement II are false.

Open solution

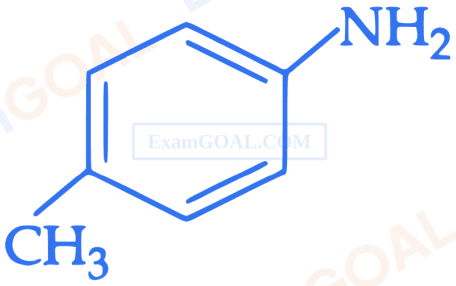
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Q11. MCQ (Single Correct Answer)

Which one of the following gives the most stable Diazonium salt?

(A) $CH_3-CH_2-CH_2-NH_2$

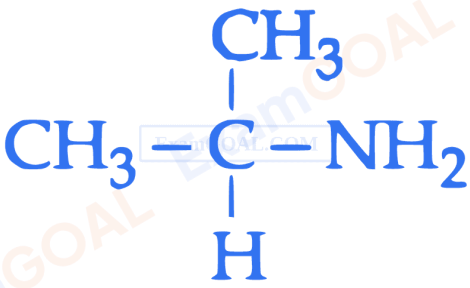
(B)



(C)



(D)



Open solution

Q12. MCQ (Single Correct Answer)

The potassium ferrocyanide solution gives a Prussian blue colour, when added to :

(A) $CoCl_3$ (B) $FeCl_2$ (C) $CoCl_2$ (D) $FeCl_3$

Open solution

Q13. MCQ (Single Correct Answer)

Number of paramagnetic oxides among the following given oxides is _____.

Li_2O, CaO, Na_2O_2, KO_2, MgO and K_2O

(A) 1 (B) 2 (C) 3 (D) 0

Open solution

Q14. MCQ (Single Correct Answer)

Identify the element for which electronic configuration in +3 oxidation state is $[Ar]3d^5$:

(A) Ru (B) Mn (C) Co (D) Fe

Open solution

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Q15. Numerical

The molar solubility of $Zn(OH)_2$ in 0.1 M NaOH solution is $x \times 10^{-18}$ M. The value of x is _____ (Nearest integer)

(Given : The solubility product of $Zn(OH)_2$ is 2×10^{-20})

Open solution

Q16. Numerical

For the reaction, $2NO_2(g) \rightleftharpoons N_2O_4(g)$, when $\Delta S = -176.0\ JK^{-1}$ and $\Delta H = -57.8\ kJ\ mol^{-1}$, the magnitude of ΔG at 298 K for the reaction is _____ $kJ\ mol^{-1}$. (Nearest integer)

Open solution

Q17. Numerical

The sum of oxidation states of two silver ions in $[Ag(NH_3)_2]\ [Ag(CN)_2]$ complex is _____.

Open solution

Q18. Numerical

The number of atoms in 8g of sodium is $x \times 10^{23}$. The value of x is _____. (Nearest integer)

[Given : $N_A = 6.02 \times 10^{23}\ mol^{-1}$

Atomic mass of Na = 23.0 u]

Open solution

Q19. Numerical

If 80 g of copper sulphate $CuSO_4 . 5H_2O$ is dissolved in deionised water to make 5L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3}$ mol L^{-1}. The value of x is _____.

[Atomic masses Cu : 63.54u, S : 32u, O : 16u, H : 1u]

Open solution

Q20. Numerical

A 50 watt bulb emits monochromatic red light of wavelength of 795 nm. The number of photons emitted per second by the bulb is $x \times 10^{20}$. The value of x is _____. [Given : $h = 6.63 \times 10^{-34}$ Js and $c = 3.0 \times 10^8\ ms^{-1}$]

Open solution

Q21. Numerical

The spin-only magnetic moment value of B_2^+ species is _____ $\times 10^{-2}$ BM. (Nearest integer)
[Given : $\sqrt{3}$ = 1.73]

Open solution

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Q22. Numerical

If the conductivity of mercury at 0°C is 1.07×10^6 S m^{-1} and the resistance of a cell containing mercury is 0.243Ω, then the cell constant of the cell is x $\times$ 10^4 m^{-1}. The value of x is _____. (Nearest integer)

Open solution

Q23. Numerical

A peptide synthesized by the reactions of one molecule each of Glycine, Leucine, Aspartic acid and Histidine will have _____ peptide linkages.

Open solution

Mathematics 23 questions

Q24. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be a continuous function. Then $\lim\limits_{x \to \frac{\pi}{4}} \frac{\int_{2}^{\sec^2 x} f(x)\,dx}{x^2 - \frac{\pi^2}{16}}$ is equal to :

(A) f (2) (B) 2f (2) (C) 2f $\left(\sqrt{2}\right)$ (D) 4f (2)

Open solution

Q25. MCQ (Single Correct Answer)

$\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$ is equal to :

(The inverse trigonometric functions take the principal values)

(A) $3\pi - 11$ (B) $4\pi - 9$ (C) $4\pi - 11$ (D) $3\pi + 1$

Open solution

Q26. MCQ (Single Correct Answer)

Consider the system of linear equations

$-x + y + 2z = 0$

$3x - ay + 5z = 1$

$2x - 2y - az = 7$

Let S_1 be the set of all a∈R for which the system is inconsistent and S_2 be the set of all a∈R for which the system has infinitely many solutions. If $n(S_1)$ and $n(S_2)$ denote the number of elements in S_1 and S_2 respectively, then

(A) $n(S_1) = 2, n(S_2) = 2$ (B) $n(S_1) = 1, n(S_2) = 0$

(C) $n(S_1) = 2, n(S_2) = 0$ (D) $n(S_1) = 0, n(S_2) = 2$

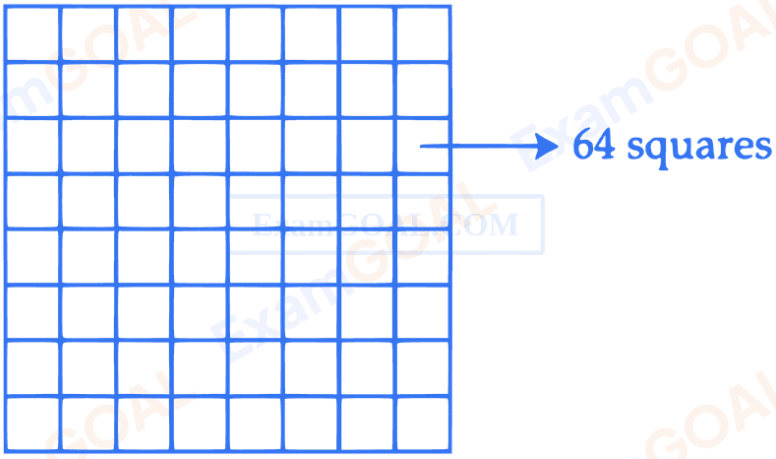
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Q27. MCQ (Single Correct Answer)

Two squares are chosen at random on a chessboard (see figure). The probability that they have a side in common is :

Open solution



(A) $\frac{2}{7}$ (B) $\frac{1}{18}$ (C) $\frac{1}{7}$ (D) $\frac{1}{9}$

Q28. MCQ (Single Correct Answer)

If y = y(x) is the solution curve of the differential equation $x^2dy + \left(y - \frac{1}{x}\right)dx = 0$; x > 0 and y(1) = 1, then $y\left(\frac{1}{2}\right)$ is equal to :

Open solution

(A) $\frac{3}{2} - \frac{1}{\sqrt{e}}$ (B) $3 + \frac{1}{\sqrt{e}}$

(C) 3 + e (D) $3 - e$

Q29. MCQ (Single Correct Answer)

The function $f(x) = x^3 - 6x^2 + ax + b$ is such that $f(2) = f(4) = 0$. Consider two statements :

Open solution

Statement 1 : there exists $x_1, x_2 \in (2, 4)$, $x_1 < x_2$, such that $f'(x_1) = -1$ and $f'(x_2) = 0$.

Statement 2 : there exists $x_3, x_4 \in (2, 4)$, $x_3 < x_4$, such that f is decreasing in $(2, x_4)$, increasing in $(x_4, 4)$ and $2f'(x_3) = \sqrt{3}f(x_4)$.

Then

(A) both Statement 1 and Statement 2 are true

(B) Statement 1 is false and Statement 2 is true

(C) both Statement 1 and Statement 2 are false

(D) Statement 1 is true and Statement 2 is false

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Q30. MCQ (Single Correct Answer)

Let $J_{n,m} = \int_0^{\frac{1}{2}} \frac{x^n}{x^m - 1} dx$, $\forall$ n > m and n, m $\in$ N. Consider a matrix $A = [a_{ij}]_{3\times 3}$ where $a_{ij} = \begin{cases} J_{6+i,3} - J_{i+3,3}, & i \le j \\ 0, & i > j \end{cases}$. Then $\left|adjA^{-1}\right|$ is :

(A) $(15)^2 \times 2^{42}$ (B) $(15)^2 \times 2^{34}$ (C) $(105)^2 \times 2^{38}$ (D) $(105)^2 \times 2^{36}$

Open solution

Q31. MCQ (Single Correct Answer)

The area, enclosed by the curves $y = \sin x + \cos x$ and $y = |\cos x - \sin x|$ and the lines $x = 0, x = \frac{\pi}{2}$, is :

(A) $2\sqrt{2}(\sqrt{2} - 1)$ (B) $2(\sqrt{2} + 1)$ (C) $4(\sqrt{2} - 1)$ (D) $2\sqrt{2}(\sqrt{2} + 1)$

Open solution

Q32. MCQ (Single Correct Answer)

The distance of line $3y - 2z - 1 = 0 = 3x - z + 4$ from the point (2, -1, 6) is :

(A) $\sqrt{26}$ (B) $2\sqrt{5}$ (C) $2\sqrt{6}$ (D) $4\sqrt{2}$

Open solution

Q33. MCQ (Single Correct Answer)

The number of pairs (a, b) of real numbers, such that whenever α is a root of the equation $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of this equation, is :

(A) 6 (B) 2 (C) 4 (D) 8

Open solution

Q34. MCQ (Single Correct Answer)

Let $P_1, P_2, \ldots\ldots, P_{15}$ be 15 points on a circle. The number of distinct triangles formed by points P_i, P_j, P_k such that $i + j + k \neq 15$, is :

(A) 12 (B) 419 (C) 443 (D) 455

Open solution

Q35. MCQ (Single Correct Answer)

The range of the function,

$f(x) = \log_{\sqrt{5}}\left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right)\right)$ is :

(A) $\left(0, \sqrt{5}\right)$ (B) $[-2, 2]$

(C) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$ (D) $[0, 2]$

Open solution

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Q36. MCQ (Single Correct Answer)

Let $a_1, a_2, \ldots\ldots\ldots, a_{21}$ be an AP such that $\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} = \frac{4}{9}$. If the sum of this AP is 189, then $a_6 a_{16}$ is equal to :

(A) 57 (B) 72 (C) 48 (D) 36

Open solution

Q37. MCQ (Single Correct Answer)

The function f(x), that satisfies the condition $f(x) = x + \int_0^{\pi/2} \sin x . \cos y \, f(y) \, dy$, is :

(A) $x + \frac{2}{3}(\pi - 2)\sin x$ (B) $x + (\pi + 2)\sin x$ (C) $x + \frac{\pi}{2}\sin x$ (D) $x + (\pi - 2)\sin x$

Open solution

Q38. Numerical

Let X be a random variable with distribution.

x	−2	−1	3	4	6
P(X = x)	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

If the mean of X is 2.3 and variance of X is σ^2, then 100 σ^2 is equal to :

Open solution

Q39. Numerical

Let $f(x) = x^6 + 2x^4 + x^3 + 2x + 3$, x ∈ R. Then the natural number n for which $\lim_{x \to 1} \frac{x^n f(1) - f(x)}{x - 1} = 44$ is _____.

Open solution

Q40. Numerical

If for the complex numbers z satisfying | z − 2 − 2i | ≤ 1, the maximum value of | 3iz + 6 | is attained at a + ib, then a + b is equal to _____.

Open solution

Q41. Numerical

Let the points of intersections of the lines x − y + 1 = 0, x − 2y + 3 = 0 and 2x − 5y + 11 = 0 are the mid points of the sides of a triangle ΔABC. Then, the area of the ΔABC is _____.

Open solution

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Q42. Numerical

Let f(x) be a polynomial of degree 3 such that
$f(k) = -\frac{2}{k}$ for k = 2, 3, 4, 5. Then the value of $52 - 10f(10)$ is equal to :

Open solution

Q43. Numerical

All the arrangements, with or without meaning, of the word FARMER are written excluding any word that has two R appearing together. The arrangements are listed serially in the alphabetic order as in the English dictionary. Then the serial number of the word FARMER in this list is _____.

Open solution

Q44. Numerical

Let $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. Let a vector $\vec{v}$ be in the plane containing $\vec{a}$ and $\vec{b}$. If $\vec{v}$ is perpendicular to the vector $3\hat{i} + 2\hat{j} - \hat{k}$ and its projection on $\vec{a}$ is 19 units, then $\left|2\vec{v}\right|^2$ is equal to _____.

Open solution

Q45. Numerical

Let [t] denote the greatest integer $\leq$ t. The number of points where the function
$f(x) = [x]\left|x^2 - 1\right| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1], x \in (-2, 2)$ is not continuous is _____.

Open solution

Q46. Numerical

A man starts walking from the point P(−3, 4), touches the x-axis at R, and then turns to reach at the point Q(0, 2). The man is walking at a constant speed. If the man reaches the point Q in the minimum time, then $50\left((PR)^2 + (RQ)^2\right)$ is equal to _____.

Open solution

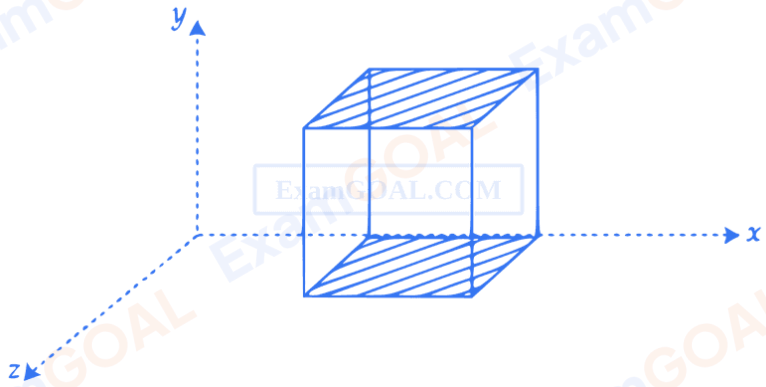
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Physics 28 questions

Q47. MCQ (Single Correct Answer)

A cube is placed inside an electric field, $\vec{E} = 150y^2\hat{j}$. The side of the cube is 0.5 m and is placed in the field as shown in the given figure. The charge inside the cube is :

Open solution

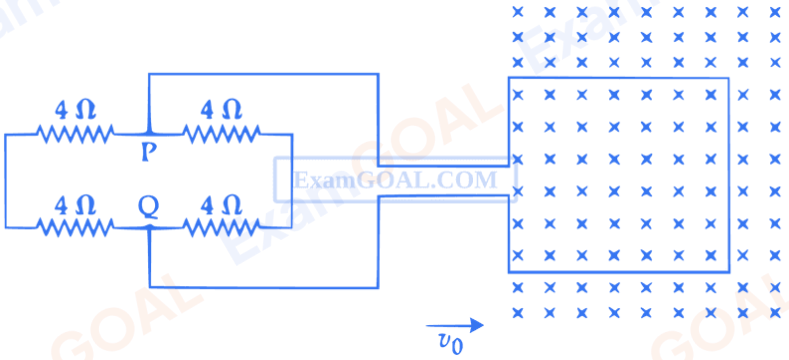


(A) 3.8×10^{-11} C (B) 8.3×10^{-11} C (C) 3.8×10^{-12} C (D) 8.3×10^{-12} C

Q48. MCQ (Single Correct Answer)

A square loop of side 20 cm and resistance 1Ω is moved towards right with a constant speed v_0. The right arm of the loop is in a uniform magnetic field of 5T. The field is perpendicular to the plane of the loop and is going into it. The loop is connected to a network of resistors each of value 4Ω. What should be the value of v_0 so that a steady current of 2 mA flows in the loop?

Open solution



(A) 1 m/s (B) 1 cm/s (C) 10^2 m/s (D) 10^{-2} cm/s

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Q49. MCQ (Single Correct Answer)

The temperature of an ideal gas in 3-dimensions is 300 K. The corresponding de-Broglie wavelength of the electron approximately at 300 K, is :

[m_e = mass of electron = 9×10^{-31} kg, h = Planck constant = 6.6×10^{-34} Js, k_B = Boltzmann constant = 1.38×10^{-23} JK^{-1}]

(A) 6.26 nm (B) 8.46 nm (C) 2.26 nm (D) 3.25 nm

Open solution

Q50. MCQ (Single Correct Answer)

A body of mass 'm' dropped from a height 'h' reaches the ground with a speed of $0.8\sqrt{gh}$. The value of work done by the air-friction is :

(A) -0.68 mgh (B) mgh (C) 1.64 mgh (D) 0.64 mgh

Open solution

Q51. MCQ (Single Correct Answer)

The ranges and heights for two projectiles projected with the same initial velocity at angles 42° and 48° with the horizontal are R_1, R_2 and H_1, H_2 respectively. Choose the correct option :

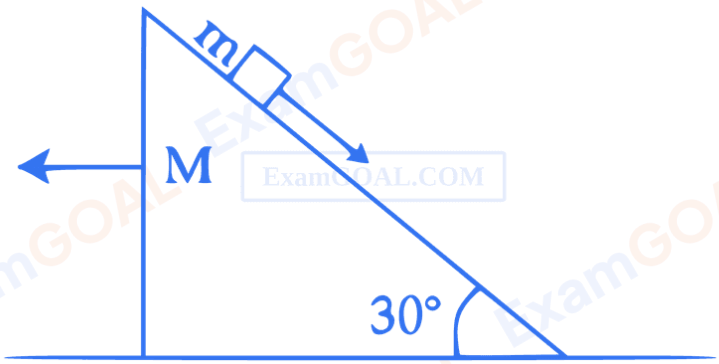
(A) $R_1 > R_2$ and $H_1 = H_2$ (B) $R_1 = R_2$ and $H_1 < H_2$

(C) $R_1 < R_2$ and $H_1 < H_2$ (D) $R_1 = R_2$ and $H_1 = H_2$

Open solution

Q52. MCQ (Single Correct Answer)

A block of mass m slides on the wooden wedge, which in turn slides backward on the horizontal surface. The acceleration of the block with respect to the wedge is : Given m = 8 kg, M = 16 kg. Assume all the surfaces shown in the figure to be frictionless.



(A) $\frac{4}{3}g$ (B) $\frac{6}{5}g$ (C) $\frac{3}{5}g$ (D) $\frac{2}{3}g$

Open solution

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Q53. MCQ (Single Correct Answer)

Open solution

Due to cold weather a 1 m water pipe of cross-sectional area 1 cm^2 is filled with ice at $-10^{\circ}C$. Resistive heating is used to melt the ice. Current of 0.5A is passed through 4 kΩ resistance. Assuming that all the heat produced is used for melting, what is the minimum time required? (Given latent heat of fusion for water/ice = 3.33×10^5 J kg^{-1}, specific heat of ice = 2×10^3 J kg^{-1} and density of ice = 10^3 kg/m^3)

(A) 0.353 s (B) 35.3 s (C) 3.53 s (D) 70.6 s

Q54. MCQ (Single Correct Answer)

Open solution

A student determined Young's Modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$. The value of g is taken to be 9.8 m/s^2, without any significant error, his observations are as follows.

Physical Quantity	Least count of the Equipment used for measurement	Observed value
Mass (M)	1 g	2 kg
Length of bar (L)	1 mm	1 m
Breadth of bar (b)	0.1 mm	4 cm
Thickness of bar (d)	0.01 mm	0.4 cm
Depression (δ)	0.01 mm	5 mm

Then the fractional error in the measurement of Y is :

(A) 0.0083 (B) 0.0155 (C) 0.155 (D) 0.083

Q55. MCQ (Single Correct Answer)

Open solution

Two resistors $R_1 = (4 \pm 0.8)\ \Omega$ and $R_2 = (4 \pm 0.4)\ \Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be :

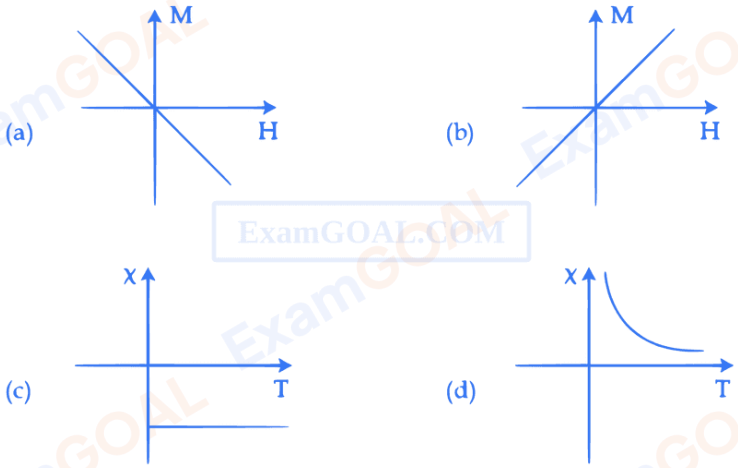
(A) $(4 \pm 0.4)\ \Omega$ (B) $(2 \pm 0.4)\ \Omega$ (C) $(2 \pm 0.3)\ \Omega$ (D) $(4 \pm 0.3)\ \Omega$

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Q56. MCQ (Single Correct Answer)

Following plots show magnetization (M) vs magnetizing field (H) and magnetic susceptibility (χ) vs temperature (T) graph :

Open solution



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Which of the following combination will be represented by a diamagnetic material?

(A) (a), (c) **(B)** (a), (d) **(C)** (b), (d) **(D)** (b), (c)

Q57. MCQ (Single Correct Answer)

A glass tumbler having inner depth of 17.5 cm is kept on a table. A student starts pouring water (μ = 4/3) into it while looking at the surface of water from the above. When he feels that the tumbler is half filled, he stops pouring water. Up to what height, the tumbler is actually filled?

Open solution

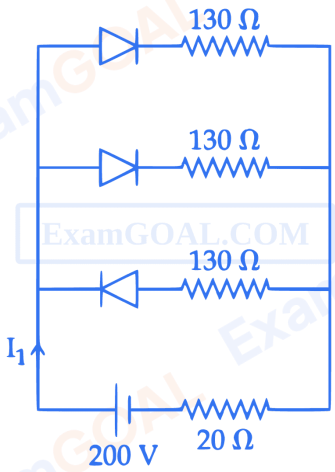
(A) 11.7 cm **(B)** 10 cm **(C)** 7.5 cm **(D)** 8.75 cm

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Q58. MCQ (Single Correct Answer)

Open solution

In the given figure, each diode has a forward bias resistance of 30Ω and infinite resistance in reverse bias. The current I_1 will be :

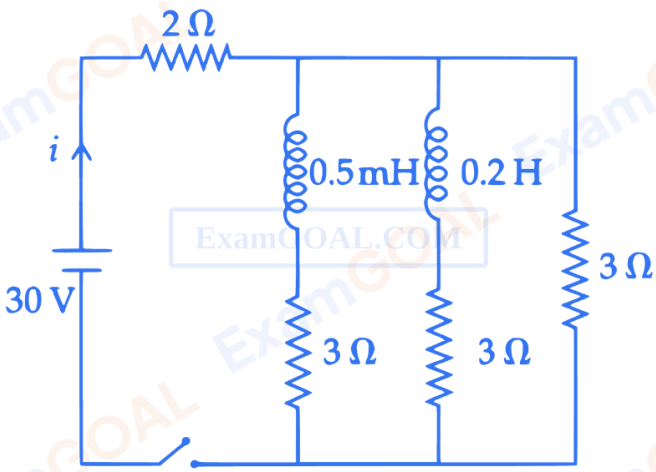


(A) 3.75 A (B) 2.35 A (C) 2 A (D) 2.73 A

Q59. MCQ (Single Correct Answer)

Open solution

For the given circuit the current i through the battery when the key is closed and the steady state has been reached is _____.



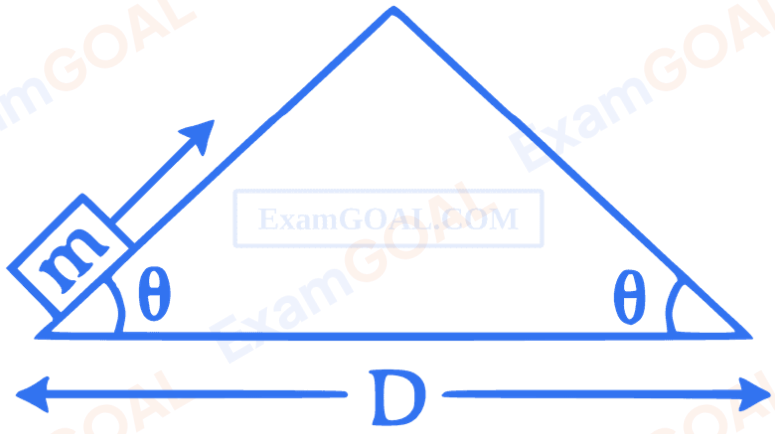
(A) 6 A (B) 25 A (C) 10 A (D) 0 A

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Q60. MCQ (Single Correct Answer)

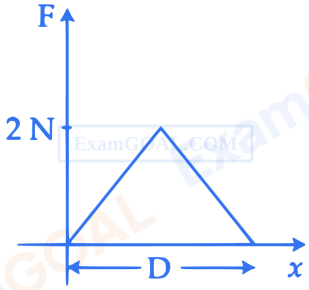
Open solution

An object of mass 'm' is being moved with a constant velocity under the action of an applied force of 2N along a frictionless surface with following surface profile.

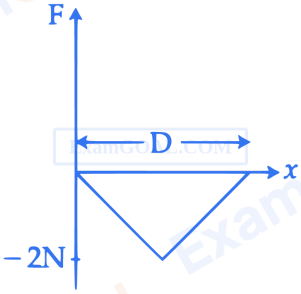


The correct applied force vs distance graph will be :

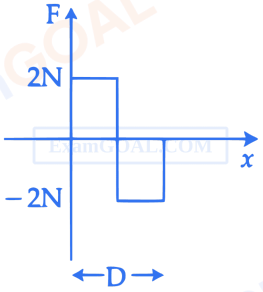
(A)



(B)

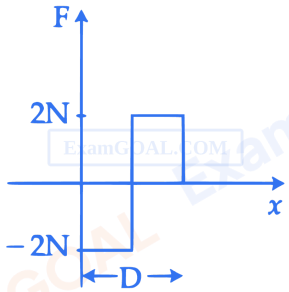


(C)



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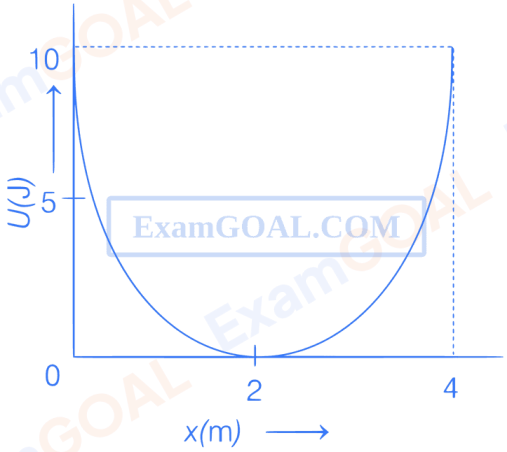
(D)



Q61. MCQ (Single Correct Answer)

A mass of 5 kg is connected to a spring. The potential energy curve of the simple harmonic motion executed by the system is shown in the figure. A simple pendulum of length 4 m has the same period of oscillation as the spring system. What is the value of acceleration due to gravity on the planet where these experiments are performed?

Open solution



(A) 10 m/s^2 (B) 5 m/s^2 (C) 4 m/s^2 (D) 9.8 m/s^2

Q62. MCQ (Single Correct Answer)

A capacitor is connected to a 20 V battery through a resistance of 10Ω. It is found that the potential difference across the capacitor rises to 2 V in 1 μs. The capacitance of the capacitor is _____ μF. Given : $\ln\left(\frac{10}{9}\right) = 0.105$

Open solution

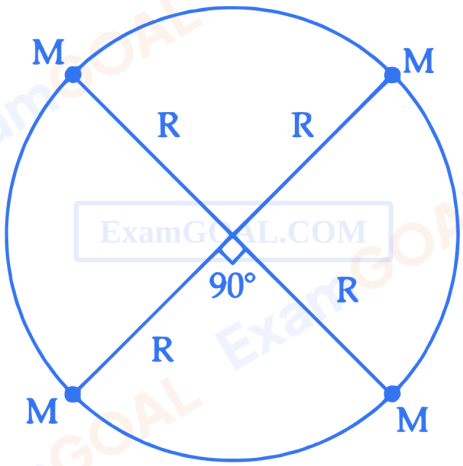
(A) 9.52 (B) 0.95 (C) 0.105 (D) 1.85

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Q63. MCQ (Single Correct Answer)

Four particles each of mass M, move along a circle of radius R under the action of their mutual gravitational attraction as shown in figure. The speed of each particle is :

Open solution



(A) $\frac{1}{2}\sqrt{\frac{GM}{R(2\sqrt{2}+1)}}$

(B) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}+1)}$

(C) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}-1)}$

(D) $\sqrt{\frac{GM}{R}}$

Q64. MCQ (Single Correct Answer)

Electric field of plane electromagnetic wave propagating through a non-magnetic medium is given by

Open solution

E = 20cos(2 $\times$ 10^{10} t $-$ 200x) V/m. The dielectric constant of the medium is equal to : (Take μ_r = 1)

(A) 9

(B) 2

(C) $\frac{1}{3}$

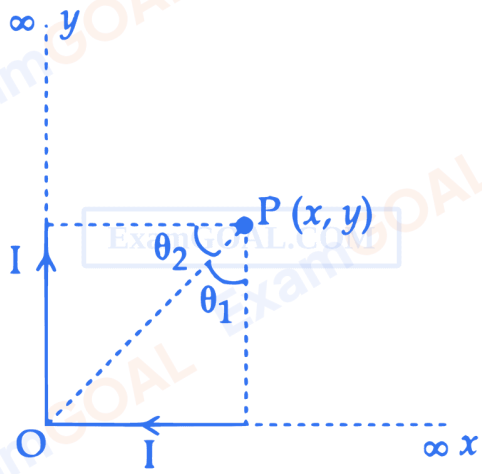
(D) 3

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Q65. MCQ (Single Correct Answer)

There are two infinitely long straight current carrying conductors and they are held at right angles to each other so that their common ends meet at the origin as shown in the figure given below. The ratio of current in both conductor is 1 : 1. The magnetic field at point P is _____.

Open solution



(A) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}+(x+y)\right]$

(B) $\frac{\mu_0 I}{4\pi xy}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(C) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}-(x+y)\right]$

(D) $\frac{\mu_0 Ixy}{4\pi}\left[\sqrt{x^2+y^2}+(x+y)\right]$

Q66. Numerical

The temperature of 3.00 mol of an ideal diatomic gas is increased by 40.0°C without changing the pressure of the gas. The molecules in the gas rotate but do not oscillate. If the ratio of change in internal energy of the gas to the amount of workdone by the gas is $\frac{x}{10}$. Then the value of x (round off to the nearest integer) is _____. (Given R = 8.31 J mol^{-1} K^{-1})

Open solution

Q67. Numerical

The width of one of the two slits in a Young's double slit experiment is three times the other slit. If the amplitude of the light coming from a slit is proportional to the slit-width, the ratio of minimum to maximum intensity in the interference pattern is x : 4 where x is _____.

Open solution

Q68. Numerical

Two satellites revolve around a planet in coplanar circular orbits in anticlockwise direction. Their period of revolutions are 1 hour and 8 hours respectively. The radius of the orbit of nearer satellite is 2×10^3 km. The angular speed of the farther satellite as observed from the nearer satellite at the instant when both the satellites are closest is $\frac{\pi}{x} rad\, h^{-1}$ where x is _____.

Open solution

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Q69. Numerical

When a body slides down from rest along a smooth inclined plane making an angle of 30° with the horizontal, it takes time T. When the same body slides down from the rest along a rough inclined plane making the same angle and through the same distance, it takes time αT, where α is a constant greater than 1. The co-efficient of friction between the body and the rough plane is $\frac{1}{\sqrt{x}}\left(\frac{\alpha^2-1}{\alpha^2}\right)$ where x = _____.

Open solution

Q70. Numerical

The average translational kinetic energy of N_2 gas molecules at .....°C becomes equal to the K.E. of an electron accelerated from rest through a potential difference of 0.1 volt. (Given k_B = 1.38 $\times 10^{-23}$ J/K) (Fill the nearest integer).

Open solution

Q71. Numerical

A uniform heating wire of resistance 36Ω is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x, where x is _____

Open solution

Q72. Numerical

A steel rod with y = 2.0 $\times 10^{11}$ Nm^{-2} and $\alpha = 10^{-5}$ $°C^{-1}$ of length 4 m and area of cross-section 10 cm^2 is heated from 0°C to 400°C without being allowed to extend. The tension produced in the rod is x $\times 10^5$ N where the value of x is _____.

Open solution

Q73. Numerical

A 2 kg steel rod of length 0.6 m is clamped on a table vertically at its lower end and is free to rotate in vertical plane. The upper end is pushed so that the rod falls under gravity, ignoring the friction due to clamping at its lower end, the speed of the free end of rod when it passes through its lowest position is _____ ms^{-1}. (Take g = 10 ms^{-2})

Open solution

Q74. Numerical

An engine is attached to a wagon through a shock absorber of length 1.5 m. The system with a total mass of 40,000 kg is moving with a speed of 72 kmh^{-1} when the brakes are applied to bring it to rest. In the process of the system being brought to rest, the spring of the shock absorber gets compressed by 1.0 m. If 90% of energy of the wagon is lost due to friction, the spring constant is _____ $\times 10^5$ N/m.

Open solution

Out of syllabus 16 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q75. MCQ (Single Correct Answer) · **Chemistry**

Water sample is called cleanest on the basis of which one of the BOD values given below :

(A) 11 ppm (B) 15 ppm (C) 3 ppm (D) 21 ppm

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Calamine and Malachite, respectively, are the ores of :

(A) Nickel and Aluminium (B) Zinc and Copper

(C) Copper and Iron (D) Aluminium and Zinc

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II :

List - I (Colloid Preparation Method)		List - II (Chemical Reaction)	
(a)	Hydrolysis	(i)	$2AuCl_3 + 3HCHO + 3H_2O \rightarrow 2Au(sol) + 3HCOOH + 6HCl$
(b)	Reduction	(ii)	$As_2O_3 + 3H_2S \rightarrow As_2S_3(sol) + 3H_2O$
(c)	Oxidation	(iii)	$SO_2 + 2H_2S \rightarrow 3S(sol) + 2H_2O$
(d)	Double Decomposition	(iv)	$FeCl_3 + 3H_2O \rightarrow Fe(OH)_3(sol) + 3HCl$

Choose the most appropriate answer from the options given below.

(A) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv) (B) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

(C) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i) (D) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

Open solution

Q78. MCQ (Single Correct Answer) · **Chemistry**

Monomer units of Dacron polymer are :

(A) ethylene glycol and phthalic acid (B) ethylene glycol and terephthalic acid

(C) glycerol and terephthalic acid (D) glycerol and phthalic acid

Open solution

Q79. MCQ (Single Correct Answer) · **Chemistry**

Hydrogen peroxide reacts with iodine in basic medium to give :

(A) IO_4^- (B) IO^- (C) I^- (D) IO_3^-

Open solution

Q80. MCQ (Single Correct Answer) · **Chemistry**

The oxide without nitrogen-nitrogen bond is :

(A) N_2O (B) N_2O_4 (C) N_2O_3 (D) N_2O_5

Open solution

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Q81. Numerical · **Chemistry**

An empty LPG cylinder weighs 14.8 kg. When full, it weighs 29.0 kg and shows a pressure of 3.47 atm. In the course of use at ambient temperature, the mass of the cylinder is reduced to 23.0 kg. The final pressure inside of the cylinder is _____ atm. (Nearest integer)

(Assume LPG to be an ideal gas)

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let the acute angle bisector of the two planes $x - 2y - 2z + 1 = 0$ and $2x - 3y - 6z + 1 = 0$ be the plane P. Then which of the following points lies on P?

(A) $\left(3, 1, -\frac{1}{2}\right)$ (B) $\left(-2, 0, -\frac{1}{2}\right)$

(C) $(0, 2, -4)$ (D) $(4, 0, -2)$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Which of the following is equivalent to the Boolean expression $p \wedge \sim q$?

(A) $\sim (q \rightarrow p)$ (B) $\sim p \rightarrow \sim q$ (C) $\sim (p \rightarrow \sim q)$ (D) $\sim (p \rightarrow q)$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

If n is the number of solutions of the equation
$2\cos x\left(4\sin\left(\frac{\pi}{4}+x\right)\sin\left(\frac{\pi}{4}-x\right)-1\right)=1, x \in [0, \pi]$ and S is the sum of all these solutions, then the ordered pair (n, S) is :

(A) $(3, 13\pi / 9)$ (B) $(2, 2\pi / 3)$ (C) $(2, 8\pi / 9)$ (D) $(3, 5\pi / 3)$

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

Consider the parabola with vertex $\left(\frac{1}{2}, \frac{3}{4}\right)$ and the directrix $y = \frac{1}{2}$. Let P be the point where the parabola meets the line $x = -\frac{1}{2}$. If the normal to the parabola at P intersects the parabola again at the point Q, then $(PQ)^2$ is equal to :

(A) $\frac{75}{8}$ (B) $\frac{125}{16}$ (C) $\frac{25}{2}$ (D) $\frac{15}{2}$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let S_n = 1 . (n − 1) + 2 . (n − 2) + 3 . (n − 3) + ... + (n − 1) . 1, $n \geq 4$.

The sum $\sum_{n=4}^{\infty}\left(\frac{2S_n}{n!}-\frac{1}{(n-2)!}\right)$ is equal to :

(A) $\frac{e-1}{3}$ (B) $\frac{e-2}{6}$ (C) $\frac{e}{3}$ (D) $\frac{e}{6}$

Open solution

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Q87. MCQ (Single Correct Answer) · **Mathematics**

Let θ be the acute angle between the tangents to the ellipse $\frac{x^2}{9}+\frac{y^2}{1}=1$ and the circle $x^2+y^2=3$ at their point of intersection in the first quadrant. Then $\tan\theta$ is equal to :

(A) $\frac{5}{2\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{4}{\sqrt{3}}$ (D) 2

Open solution

Q88. Numerical · **Mathematics**

If the sum of the coefficients in the expansion of $(x+y)^n$ is 4096, then the greatest coefficient in the expansion is _____.

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

The half life period of radioactive element x is same as the mean life time of another radioactive element y. Initially they have the same number of atoms. Then :

(A) x-will decay faster than y.

(B) y-will decay faster than x.

(C) x and y have same decay rate initially and later on different decay rate.

(D) x and y decay at the same rate always.

Open solution

Q90. Numerical · **Physics**

A carrier wave with amplitude of 250 V is amplitude modulated by a sinusoidal base band signal of amplitude 150V. The ratio of minimum amplitude to maximum amplitude for the amplitude modulated wave is 50 : x, then value of x is _____.

Open solution

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Answer Key

Chemistry

1. **B**	2. **A**	3. **A**	4. **C**	5. **D**
6. **A, D**	7. **D**	8. **B**	9. **D**	10. **B**
11. **B**	12. **D**	13. **A**	14. **D**	15. **2**
16. **5**	17. **2**	18. **2**	19. **64**	20. **2**
21. **173**	22. **26**	23. **3**		

Mathematics

24. **B**	25. **C**	26. **C**	27. **B**	28. **D**
29. **A**	30. **C**	31. **A**	32. **C**	33. **A**
34. **C**	35. **D**	36. **B**	37. **D**	38. **781**
39. **7**	40. **5**	41. **6**	42. **26**	43. **77**
44. **1494**	45. **2**	46. **1250**		

Physics

47. **B**	48. **B**	49. **A**	50. **A**	51. **B**
52. **D**	53. **B**	54. **B**	55. **C**	56. **A**
57. **B**	58. **C**	59. **C**	60. **C**	61. **C**
62. **B**	63. **B**	64. **A**	65. **A**	66. **25**
67. **1**	68. **3**	69. **3**	70. **500**	71. **4**
72. **8**	73. **6**	74. **16**		

Out of syllabus

75. **C**	76. **B**	77. **B**	78. **B**	79. **C**
80. **D**	81. **2**	82. **B**	83. **D**	84. **A**
85. **B**	86. **A**	87. **B**	88. **924**	89. **B**
90. **200**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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